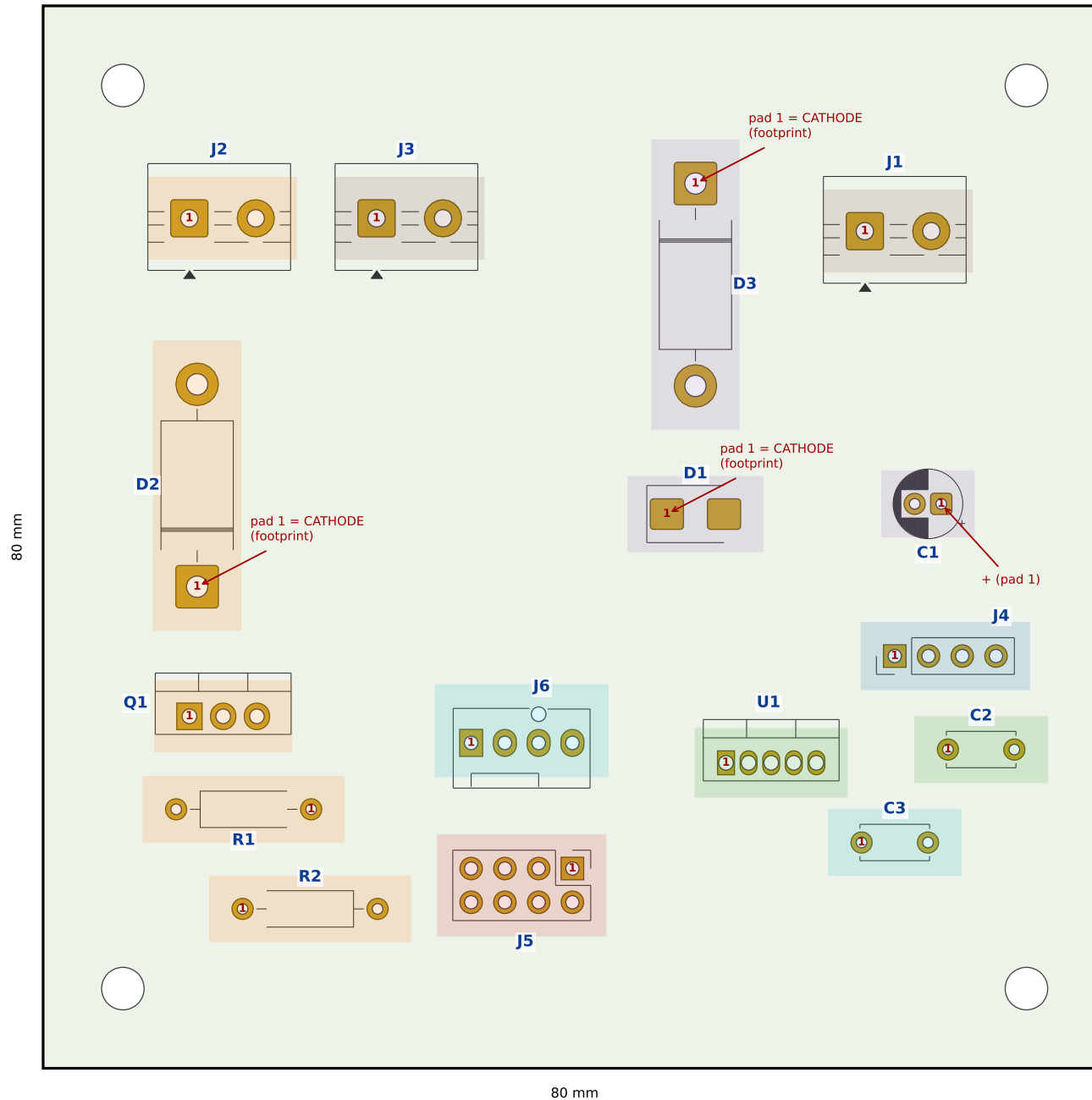


OSBAMS Rev.2 controller PCB - assembly drawing - TOP view

Derived from 'OSBAMS PCB.kicad_pcb' by tools/mfg - NOT a KiCad plot. Pad '1' markers in red; square pads = pin 1.



Highlighted sections

Controller interface (NUCLEO-L476RG carries the STM32L476RG - NOT on this PCB): J5

INA228 module header (INA228 is on an external breakout): J4

Temperature interface (TC74): U1, C2

Relay (contactor-coil) driver: Q1, R1, R2, D2, J2

Protection (reverse-polarity, TVS, bulk cap): D1, D3, C1

Power / E-stop connectors: J1, J3

UART / Pi interface: J6, C3

STM32L476RG: NOT on this PCB - it is on the NUCLEO-L476RG module that plugs into J5.

INA228: NOT on this PCB - external Adafruit breakout plugs into J4 (3V3/GND/SCL/SDA).

Programming header: NONE on this PCB (ST-LINK is on the Nucleo module).

Test points: NONE on this PCB (see Test Point Map).

Analog measurement section: none beyond the I2C sensors (no ADC divider on this PCB).

! D1, D2, D3: 'pad 1' is the CATHODE per the KiCad footprints but the schematic

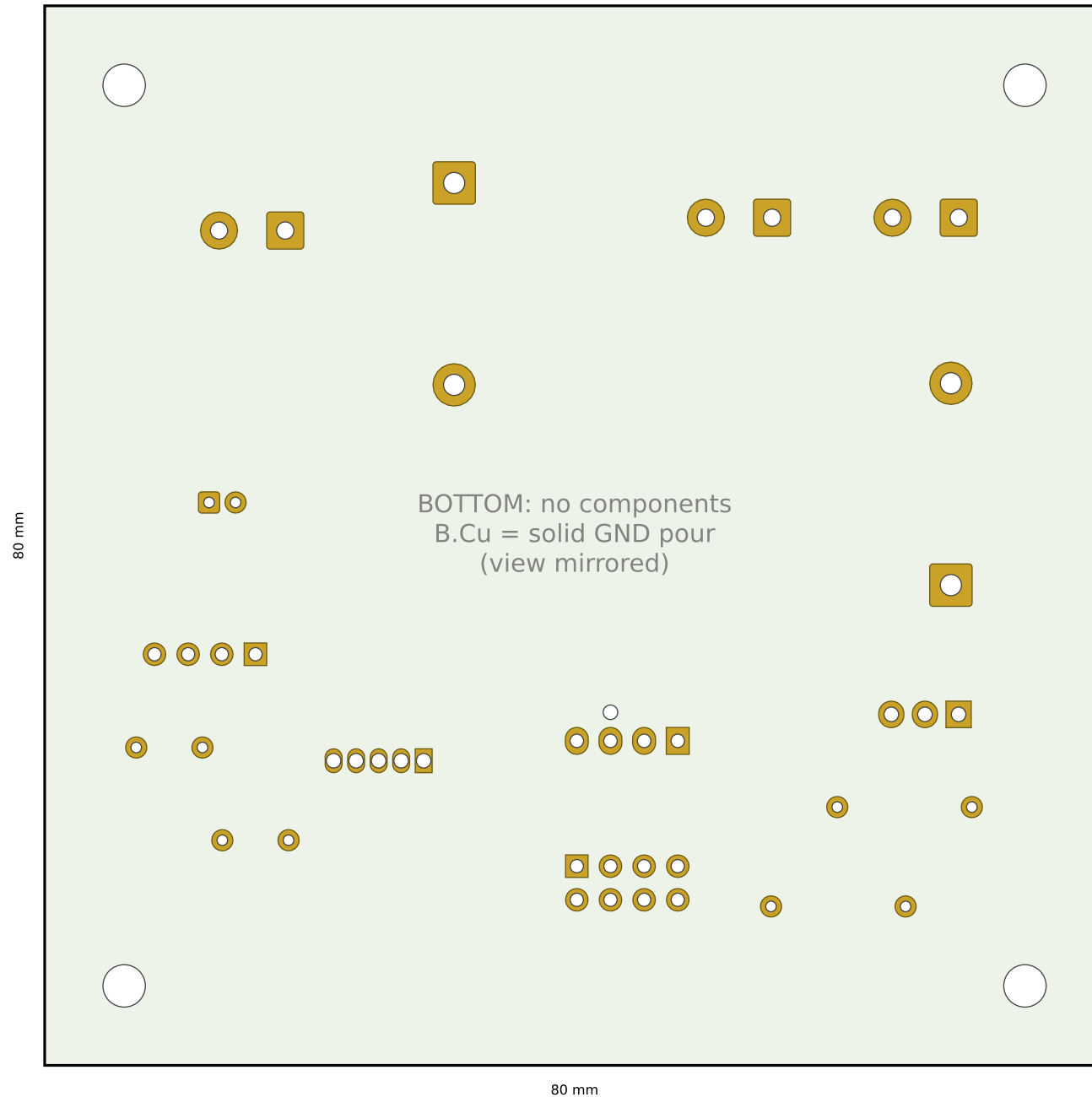
! symbol has pin 1 = ANODE -> polarity mismatch. Do NOT assemble until resolved.

Square pad / red '1' = pin 1 of connectors. C1 electrolytic: + = pad 1.

Board carries only 12 V coil current, NOT the battery discharge current.

OSBAMS Rev.2 controller PCB - assembly drawing - BOTTOM view (mirrored)

Derived from 'OSBAMS PCB.kicad_pcb' by tools/mfg - NOT a KiCad plot. Pad '1' markers in red; square pads = pin 1.



Bottom side: no components.

B.Cu is a solid GND pour (thermal-relief connections to GND pads); all signal tracks are on F.Cu.

No bottom silkscreen, no bottom paste.

OSBAMS Rev.2 controller PCB - polarity, orientation and special assembly notes

Generated from the KiCad design files. Pin-1 and polarity data below are read from the PCB netlist/footprints.

Special assembly notes

Scope: this PCB does not carry the battery discharge current. It is a 12 V coil-driver / sensor-interface carrier board. Battery fuse, disconnect, relay contacts, shunt wiring, XT60 power connector and the 6060B connection are external.

Modules are not assembled by the fab: the NUCLEO-L476RG (STM32L476RG) and the INA228 breakout are plugged in by the customer.

SMT: only D1 (SMBJ15A, DO-214AA/SMB, top side). Everything else is through-hole (TO-220 x2, axial parts, radial parts, terminal blocks, pin headers).

Diodes D1, D2, D3 - HOLD: the KiCad footprints put the cathode on pad 1; the schematic symbol puts the anode on pin 1. As drawn, the PCB connects each diode's cathode to the wrong net. Resolve in KiCad, regenerate, then assemble.

C1 electrolytic: + lead in pad 1 (square/rounded pad, silkscreen '+').

Q1 (TO-220-3): pins G-D-S left to right as marked; **U1 (TC74, TO-220-5):** pin 1 = NC at the square pad.

J5 (2x4 header): schematic calls it keyed, the footprint is not; mark orientation with pin-1 square pad. **J6** uses a polarised fan-header footprint.

Do not substitute U1, Q1, D1, D2, D3, R2 (gate pull-down) or the external STM32/INA228 modules without written approval.

Mounting holes MH1-MH4: 3.2 mm NPTH for M3 hardware (not supplied).

Pin / polarity map

Ref	Pad	Net	Notes
C1	1	+12V	+ (positive)
C1	2	GND	- (negative)
C2	1	+3V3	1_1
C2	2	GND	2_2
C3	1	+3V3	1_1
C3	2	GND	2_2
D1	1	GND	pad 1 = CATHODE (footprint) - on GND [MISMATCH]
D1	2	+12V	pad 2 = anode (footprint) - on +12V [MISMATCH]
D2	1	COIL_SW	pad 1 = CATHODE (footprint) - on COIL_SW [MISMATCH]
D2	2	COIL_V	pad 2 = anode (footprint) - on COIL_V [MISMATCH]
D3	1	12V_IN	pad 1 = CATHODE (footprint) - on 12V_IN [MISMATCH]
D3	2	+12V	pad 2 = anode (footprint) - on +12V [MISMATCH]
J1	1	12V_IN	12V_IN (+12 V input, before D3)
J1	2	GND	GND
J2	1	COIL_V	COIL_V (coil +, via E-stop loop)
J2	2	COIL_SW	COIL_SW (coil -, MOSFET drain)
J3	1	+12V	+12V (E-stop loop in)
J3	2	COIL_V	COIL_V (E-stop loop out)
J4	1	+3V3	+3V3
J4	2	GND	GND
J4	3	SCL	SCL
J4	4	SDA	SDA
J5	1	+3V3	P1_1
J5	2	GND	P2_2
J5	3	SDA	P3_3
J5	4	SCL	P4_4

Ref	Pad	Net	Notes
J5	5	GPIO_GATE	P5_5
J5	6	UART_TX	P6_6
J5	7	UART_RX	P7_7
J5	8	SPARE_GPIO	P8_8
J6	1	+3V3	+3V3
J6	2	GND	GND
J6	3	UART_TX	UART_TX
J6	4	UART_RX	UART_RX
Q1	1	GATE	Gate
Q1	2	COIL_SW	Drain
Q1	3	GND	Source
R1	1	GPIO_GATE	1_1
R1	2	GATE	2_2
R2	1	GATE	1_1
R2	2	GND	2_2
U1	1	unconnected-(U1-NC-Pad 1)	NC
U1	2	SDA	SDA
U1	3	GND	GND
U1	4	SCL	SCLK
U1	5	+3V3	VDD (3.3 V)